



Locky Ransomware Clustering Case Study

A case study showing Bitcoin address clustering with the common-input ownership heuristic.

Case focus

This case study begins with a Bitcoin address linked to the Locky ransomware family from a published SANS case study. Public blockchain data from Blockstream was then used to trace related transactions.

The common-input ownership heuristic (CIOH) was used to look for addresses that may be controlled by the same entity. The wider 30-address cluster was also cross-checked against the public Ransomwhere dataset, where those addresses were associated with Locky.

Later wallet activity shown in this case is simply following where the Bitcoin moved next, because ownership becomes less certain once funds leave the original cluster.

How the analysis works

Common-input ownership heuristic (CIOH)

CIOH groups Bitcoin addresses that are used together as inputs in the same transaction. The idea is that spending from each input address usually requires control of that address's private key.

This makes CIOH useful for blockchain clustering, but it is not perfect proof of shared ownership. Exchange wallets, collaborative transactions, shared services, and privacy tools can weaken this assumption.

Locky-associated inputs	Seed contribution	BTC combined	Main output	Second output
30	3.0000 BTC	80.0134 BTC	80.0000 BTC	0.0100030...

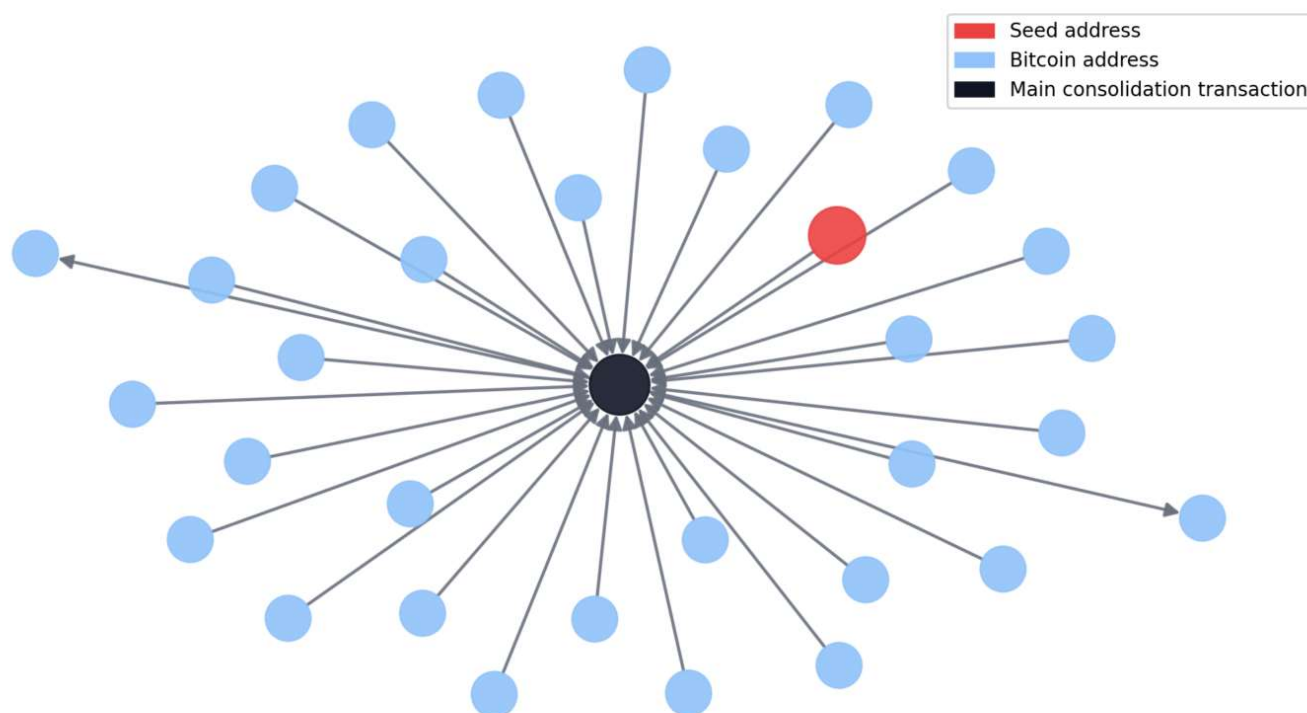
What the transaction shows

Thirty Locky-associated addresses were used together as inputs in one Bitcoin transaction. Under the common-input ownership heuristic, this supports the idea that the input addresses may have been controlled by the same wallet or operator group.

1. Consolidation flow

This graph is the main visual story: many Locky-associated input addresses are consolidated, then almost all BTC is sent to one 80 BTC output address.

Locky BTC consolidation flow

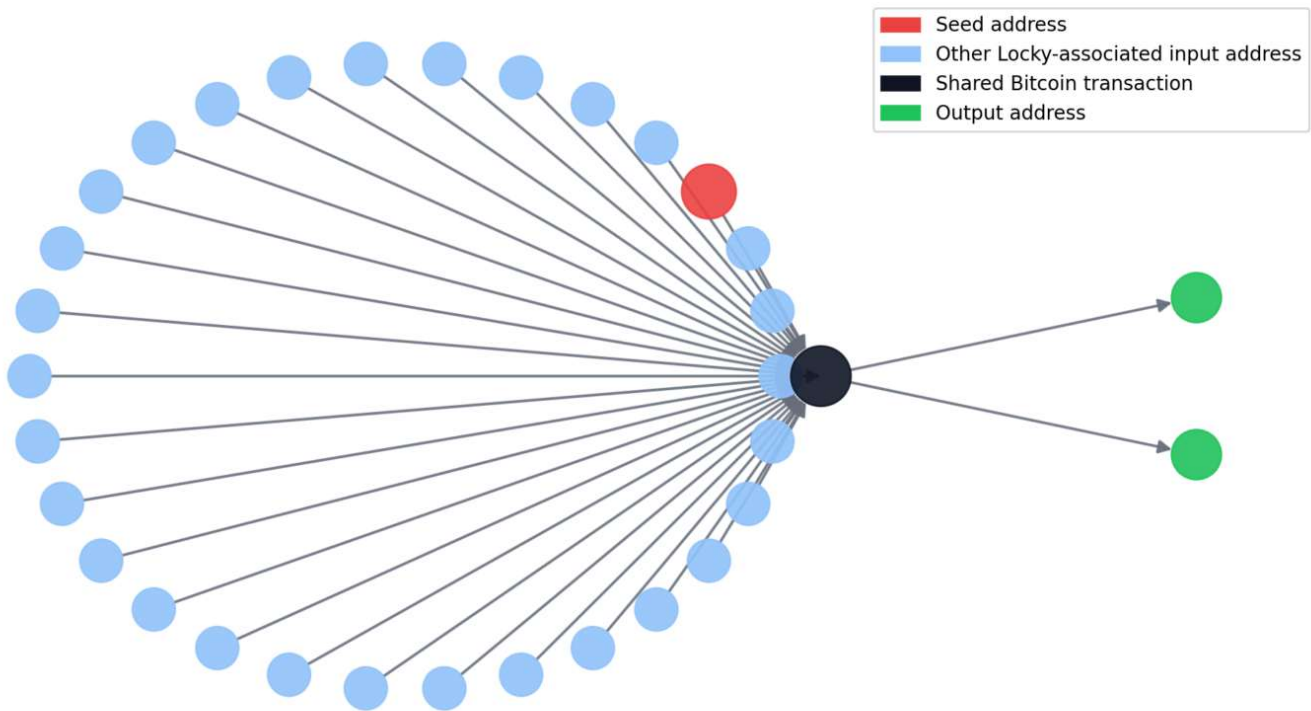


2. Common-input ownership heuristic evidence

This graph explains why the consolidation is treated as a clustering example: the seed address and the other Locky-associated input addresses were spent together in the same transaction.

How funds reached the seed address: Before the main consolidation event, the seed address received 3 BTC from a transaction funded by three separate input addresses. Although this earlier transaction also involves multiple inputs, the key CIOH example in this case is the later transaction where 30 Locky-associated addresses were co-spent together, creating a much stronger clustering signal.

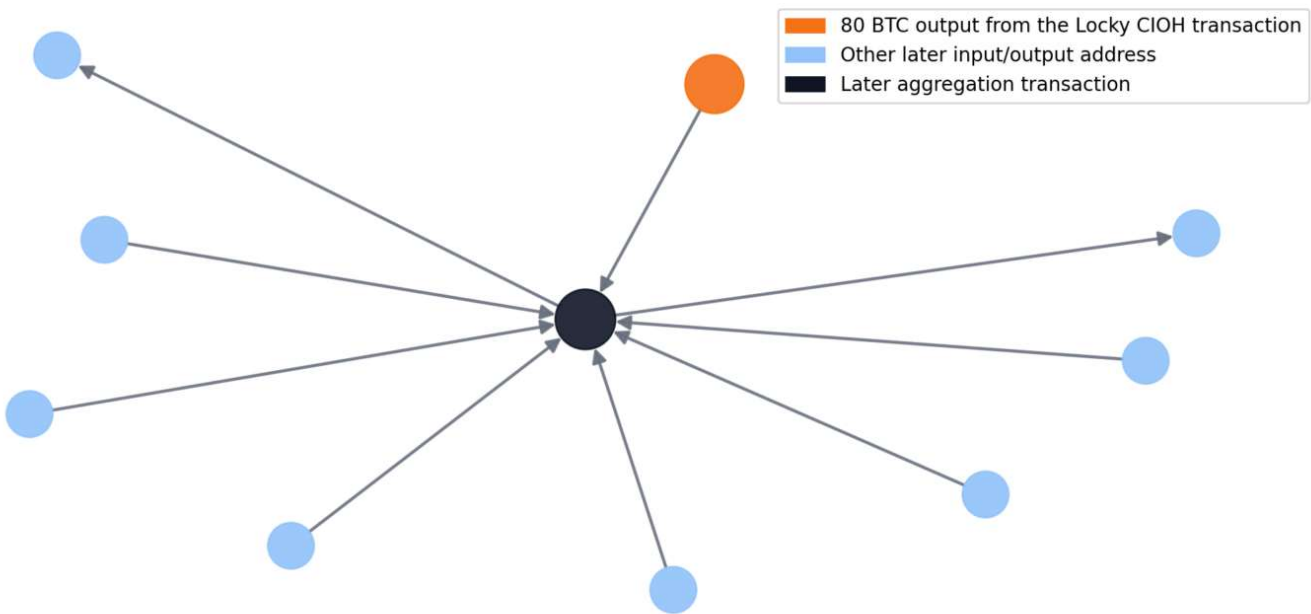
Common-input ownership heuristic: 30 Locky-associated inputs spent together



3. Where did the 80 BTC go next?

The 80 BTC output address later appeared as an input in a larger aggregation transaction. These later addresses are not labelled in the ransomware dataset, so this section is a trace, not a new attribution claim.

Follow-on trace: 80 BTC output later joins a 512.999 BTC aggregation



Follow-on transaction inputs

7

BTC combined later

512.99958568 BTC

Largest later output

500.00000000 BTC

Tracing beyond the labelled Locky cluster

Step	Date	From address	To address	BTC moved	Interpretation
1	29 Aug 2016	30 Locky-associated addresses	1Q1ifiCyTtoYsrq2MQjZqpHSFREDTteE8E	80 BTC	Initial Locky-Cl
2	06 Sep 2016	1Q1ifiCyTtoYsrq2MQjZqpHSFREDTteE8E (+ 6 other inputs)	16YhEbMcksa6zgf2rjcAUWy7fZ9TkgFNXF	500 BTC	Funds merge
3	Unknown	16YhEbMcksa6zgf2rjcAUWy7fZ9TkgFNXF	17e2WMCfReEsRev8nmC9SdbjDYfXjkGcMM	100 BTC	First redistrib
4	06 Sep 2016	16YhEbMcksa6zgf2rjcAUWy7fZ9TkgFNXF	17VSgeazX2nz4kfEjG5o5Tt6TUqHkWXs7U	399.99992101 BTC	Remaining fu
5	06 Sep 2016	17VSgeazX2nz4kfEjG5o5Tt6TUqHkWXs7U	1GMh6Qq79ZR4AQEeTMWRz2s1uP2NJ2YAeg	199.99984202 BTC	Further redisi
6	06 Sep 2016	17VSgeazX2nz4kfEjG5o5Tt6TUqHkWXs7U	1NKi9AK5R3Y8DQQgrwneCzDz5QkpUkLjHJ	200 BTC	Further redisi

Beyond the original Locky-associated address cluster, blockchain movement remains visible but attribution becomes less certain because the later wallets were not labelled in the ransomware dataset.

Follow-on trace note

The address [1Q1ifiCyTtoYsrq2MQjZqpHSFREDTteE8E](#) received the 80 BTC output from the Locky CIOH transaction. It was later spent together with six other inputs in a larger transaction totalling 512.99958568 BTC. The largest later output was 500 BTC.

The funds can still be followed on the blockchain, but it becomes harder to tell who controls the later wallets.

▼ Show follow-on input and output tables

Follow-on input addresses

Role	Address	BTC
Other follow-on input	1GQzjng6yzJLEhZRgsLvWKwvdsE9rqSDHe	
Other follow-on input	1MSApyGxayg8dJ1n3DGAsWeZTwEBpthJZB	
80 BTC output address from Locky transaction	1Q1ifiCyTtoYsrq2MQjZqpHSFREDTteE8E	
Other follow-on input	13YF7J3xixiT6Lx85shtQdq7Kuf9rEpsfw	
Other follow-on input	12oDRxwJg2FwM2We8WkKoqJMMq8jYcw2Vk	
Other follow-on input	1Pu8ighTU2SJ62r2iqfiJ11KKAYGoABX2B	
Other follow-on input	12WzkCue9ADXzKD4WXxQFsCTnv4fomMwfo	

Follow-on output addresses

Address	BTC
16YhEbMcksa6zgf2rjcAUWy7fZ9TkgFNXF	
17JuZi7GfqPdARPL341WUuy1gqqkLzy5AX	

Transaction summary

Step	What happened	BTC
1. Starting point	The seed address is one of 30 Locky-associated inputs in the transaction.	3.0000 BTC fr
2. CIOH evidence	The 30 input addresses were spent together in one transaction.	80.0134 BTC o
3. Main output	Most of the combined BTC was sent to output address 1Q1ifiCyTtoYsrq2MQjZqpHSFREDTteE8E.	80.0000 BTC
4. Second output	A very small amount was sent to output address 12p2CcaDixL2FCMBzxzfMhPwufMohDbTmH.	0.01000307 B1

▼ Show full input and output address tables

Input addresses

Role	Address	BTC
Other Locky-associated input	1Jb7zNP86WHNhJTFdWsX9mKu3akN691cgU	
Other Locky-associated input	17iuC3ca7RKnCuh7HJ1zUXfg4SsLU2suCr	
Other Locky-associated input	1TvGxNJSk7rRZf7p1bb7hHs2ZDiJLiHLG	
Other Locky-associated input	16Akt4JeRRKySzKDBRJEEweBJ8JRpUp5dE	
Other Locky-associated input	1DiMM75HqhsJDfnLGyRJMfNMDPvgkZ4t6p	
Other Locky-associated input	1PWxRF5EpAuP88Dpu6fYPfv5rmaYPqpB7i	
Other Locky-associated input	18zV1WGWCBqb632n4dPXWjUJTHymxFc15Z	
Other Locky-associated input	1LegLVX1MM4UPoP4qxkrtcwnFpJk4pHYBH	
Other Locky-associated input	1KhYxWq9r9EVkiLnLqd1wPACi6xeCaNf8C	
Other Locky-associated input	1px2pnCAfc1si5Zs8Dj5y5wqQ7otiNcyl	

Output addresses

Address	BTC
1Q1ifiCyTtoYsrq2MQjZqpHSFREDTteE8E	
12p2CcaDixL2FCMBzxzfMhPwufMohDbTmH	

References

▼ Show references and data sources

Blockstream (n.d.) *Blockstream API documentation*. Available at: <https://blockstream.info/explorer-api> (Accessed: 24 May 2026).

Cable J (2024) *Ransomwhere: A Crowdsourced Ransomware Payment Dataset (1.1.0) [Data set]*. Available at: <https://doi.org/10.5281/zenodo.6512122> (Accessed: 24 May 2026).

SANS Institute (2017) *Tracking Bitcoin Transactions on the Blockchain - SANS DFIR Summit 2017*. Available at: <https://www.youtube.com/watch?v=1iwsouV8ouQ> (Accessed: 27 February 2026).